

Seminar 1. Bayesian Reasoning: Solutions

Bayesian Methods

1. A medical test finds a serious disease in a patient. The test is 99% accurate: if the patient has the disease, the test is positive with probability 99%; if the patient does not have the disease, the test is negative with probability 99%. The disease is rare: only one person in 10 000 has it. Find the probability that the patient really has the disease.

Interpretation. Let $D \in \{0, 1\}$ show whether the patient has the disease ($D = 1$). Let $T \in \{0, 1\}$ be the test result ($T = 1$ is a positive result). The problem gives three values:

$$p(D = 1) = 10^{-4}, \quad p(T = 1 \mid D = 1) = 0.99, \quad p(T = 0 \mid D = 0) = 0.99.$$

Find $p(D = 1 \mid T = 1)$.

Idea. The disease causes the test result, so the causal graph is $D \rightarrow T$. The graph gives the joint distribution as a product of two factors:

$$p(D, T) = p(D) p(T \mid D).$$

The problem gives both factors: the distribution of the source node D and the conditional distribution on the edge $D \rightarrow T$. Thus the joint distribution is known. From the joint distribution, you can get any other probability with two operations:

- A marginal probability is the sum of the joint probability over the other variables:

$$p(T) = \sum_D p(D, T).$$

For continuous variables x and y , the sum is an integral: $p(y) = \int p(x, y) dx$.

- A conditional probability is a joint probability divided by a marginal probability:

$$p(D \mid T) = \frac{p(D, T)}{p(T)}.$$

These two operations together give Bayes' theorem:

$$p(D \mid T) = \frac{p(D, T)}{p(T)} = \frac{p(D, T)}{\sum_{D'} p(D', T)} = \frac{p(D) p(T \mid D)}{\sum_{D'} p(D') p(T \mid D')}.$$

Solution.

Step 1. Find the missing probabilities.

The probabilities of the two values of a variable add to 1. Thus

$$\begin{aligned} p(D = 0) &= 1 - p(D = 1) = 1 - 0.0001 = 0.9999, \\ p(T = 1 \mid D = 0) &= 1 - p(T = 0 \mid D = 0) = 1 - 0.99 = 0.01. \end{aligned}$$

The second value is the probability of a false positive result.

Step 2. Find the joint probabilities of a positive result.

Use $p(D, T) = p(D) p(T \mid D)$:

$$\begin{aligned} p(D = 1, T = 1) &= p(D = 1) p(T = 1 \mid D = 1) = 0.0001 \cdot 0.99 = 0.000099, \\ p(D = 0, T = 1) &= p(D = 0) p(T = 1 \mid D = 0) = 0.9999 \cdot 0.01 = 0.009999. \end{aligned}$$

Step 3. Find the marginal probability of a positive result.

Sum the joint probabilities over D :

$$p(T = 1) = p(D = 1, T = 1) + p(D = 0, T = 1) = 0.000099 + 0.009999 = 0.010098.$$

Step 4. Divide the joint probability by the marginal probability.

$$p(D = 1 \mid T = 1) = \frac{p(D = 1, T = 1)}{p(T = 1)} = \frac{0.000099}{0.010098} = \frac{99}{10098} = \frac{99}{99 \cdot 102} = \frac{1}{102} \approx 0.0098.$$

The patient has the disease with a probability of approximately 1%.

Conclusion. A positive result increases the probability of the disease from 0.01% to approximately 1%. But the patient is still healthy with a probability of 99%. The cause is the low prior probability. In 10 000 people, the test finds approximately $10\,000 \cdot 0.000099 \approx 1$ sick person and $10\,000 \cdot 0.009999 \approx 100$ healthy people. For this reason, most positive results of a mass test for a rare disease are false.

A second independent test gives more information. After the first positive result, the probability of the disease is $1/102$. This value is the prior probability for the second test. If the second result is also positive, then

$$p(D = 1 \mid T_1 = 1, T_2 = 1) = \frac{\frac{1}{102} \cdot 0.99}{\frac{1}{102} \cdot 0.99 + \frac{101}{102} \cdot 0.01} = \frac{0.99}{0.99 + 1.01} = 0.495.$$

2. Consider the following probabilistic model. At a seminar, a student answers badly ($A = 1$) or well ($A = 0$). This depends on whether the student has depression ($D = 1$ or $D = 0$) and whether the student went to a party the night before ($P = 1$ or $P = 0$). The party can also give the student a headache ($H = 1$). A bad answer makes the teacher unhappy ($T = 1$). Figure 1 shows the causal relations in the model. The probabilities are:

$p(A = 1 \mid P, D) \mid P \quad D$	$p(H = 1 \mid P) \mid P$	$p(T = 1 \mid A) \mid A$
0.999	0.9	0.95
0.9	0.2	0.5
0.9		
0.01		

$$p(P = 1) = 0.2, \quad p(D = 1) = 0.4.$$

Find $p(P = 1 \mid H = 1)$, $p(P = 1 \mid T = 1)$, and $p(P = 1 \mid T = 1, H = 1)$.

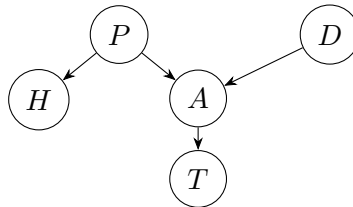


Figure 1: Causal relations for Problem 2

Idea. This problem uses the method of Problem 1 on a larger graph. The graph in Figure 1 gives the joint distribution as a product of one factor for each node. Each factor is the distribution of the node given its parents:

$$p(P, D, A, H, T) = p(P) p(D) p(A \mid P, D) p(H \mid P) p(T \mid A).$$

The tables give all five factors. Thus the joint distribution is known, and you can get any probability from it. To find the probability of P given an observation, move through the graph in two directions:

- (a) Along the edges, from P to the observed variable. Sum over each unobserved variable on the path, for example $p(T | P) = \sum_A p(T | A) p(A | P)$.
- (b) Against the edges, from the observed variable back to P . Use Bayes' theorem, for example $p(P | T) = p(P) p(T | P) / p(T)$.

Each factor $p(\text{node} | \text{parents})$ sums to 1 over the values of the node. Thus, when you sum the joint distribution over a variable that is not a parent of other variables, its factor disappears.

Solution.

Step 1. Find the missing probabilities.

The probabilities of the two values of a variable add to 1:

$$p(P = 0) = 1 - 0.2 = 0.8, \quad p(D = 0) = 1 - 0.4 = 0.6.$$

Step 2. Show that P and D are independent.

Sum the joint distribution over T , then H , then A . Each sum removes one factor, because the factor sums to 1:

$$p(P, D) = \sum_A \sum_H \sum_T p(P, D, A, H, T) = p(P) p(D).$$

Thus $p(D | P) = p(P, D) / p(P) = p(D)$.

Step 3. Write $p(A | P)$ as a sum over D .

The depression D is not observed. Sum over D , then use the definition of a conditional probability, then use Step 2:

$$p(A | P) = \sum_D p(A, D | P) = \sum_D p(A | P, D) p(D | P) = \sum_D p(A | P, D) p(D).$$

Step 4. Calculate $p(A = 1 | P)$.

Substitute the values from the tables and from Step 1:

$$\begin{aligned} p(A = 1 | P = 1) &= 0.999 \cdot 0.4 + 0.9 \cdot 0.6 = 0.3996 + 0.54 = 0.9396, \\ p(A = 1 | P = 0) &= 0.9 \cdot 0.4 + 0.01 \cdot 0.6 = 0.36 + 0.006 = 0.366. \end{aligned}$$

Thus $p(A = 0 | P = 1) = 1 - 0.9396 = 0.0604$ and $p(A = 0 | P = 0) = 1 - 0.366 = 0.634$.

Step 5. Show that $p(T | A, P) = p(T | A)$.

Sum the joint distribution over H and D . Then do the same without T :

$$\begin{aligned} p(P, A, T) &= p(P) p(T | A) \sum_D p(D) p(A | P, D), \\ p(P, A) &= p(P) \sum_D p(D) p(A | P, D). \end{aligned}$$

Divide the first line by the second line: $p(T | A, P) = p(P, A, T) / p(P, A) = p(T | A)$.

Step 6. Write $p(T | P)$ as a sum over A .

The answer A is not observed. Sum over A , then use the definition of a conditional probability, then use Step 5:

$$p(T | P) = \sum_A p(T, A | P) = \sum_A p(T | A, P) p(A | P) = \sum_A p(T | A) p(A | P).$$

Step 7. Calculate $p(T = 1 | P)$.

Substitute the values from the table and from Step 4:

$$\begin{aligned} p(T = 1 | P = 1) &= 0.95 \cdot 0.9396 + 0.5 \cdot 0.0604 = 0.89262 + 0.0302 = 0.92282, \\ p(T = 1 | P = 0) &= 0.95 \cdot 0.366 + 0.5 \cdot 0.634 = 0.3477 + 0.317 = 0.6647. \end{aligned}$$

Step 8. Find $p(P = 1 | H = 1)$.

Sum the joint distribution over T , A , and D . Each sum removes one factor, so $p(P, H) = p(P)p(H | P)$. Calculate the joint probabilities:

$$\begin{aligned} p(P = 1, H = 1) &= 0.2 \cdot 0.9 = 0.18, \\ p(P = 0, H = 1) &= 0.8 \cdot 0.2 = 0.16. \end{aligned}$$

Sum them over P : $p(H = 1) = 0.18 + 0.16 = 0.34$. Divide:

$$p(P = 1 | H = 1) = \frac{p(P = 1, H = 1)}{p(H = 1)} = \frac{0.18}{0.34} = \frac{9}{17} \approx 0.529.$$

Step 9. Find $p(P = 1 | T = 1)$.

Use $p(P, T) = p(P)p(T | P)$ and the values from Step 7:

$$\begin{aligned} p(P = 1, T = 1) &= 0.2 \cdot 0.92282 = 0.184564, \\ p(P = 0, T = 1) &= 0.8 \cdot 0.6647 = 0.53176. \end{aligned}$$

Sum them over P : $p(T = 1) = 0.184564 + 0.53176 = 0.716324$. Divide:

$$p(P = 1 | T = 1) = \frac{p(P = 1, T = 1)}{p(T = 1)} = \frac{0.184564}{0.716324} \approx 0.258.$$

Step 10. Show that H and T are independent given P .

Sum the joint distribution over the unobserved variables D and A :

$$\begin{aligned} p(P, H, T) &= \sum_A \sum_D p(P)p(D)p(A | P, D)p(H | P)p(T | A) \\ &= p(P)p(H | P) \sum_A p(T | A) \sum_D p(D)p(A | P, D) \quad (\text{move factors out of the sums}) \\ &= p(P)p(H | P) \sum_A p(T | A)p(A | P) \quad (\text{Step 3}) \\ &= p(P)p(H | P)p(T | P) \quad (\text{Step 6}). \end{aligned}$$

Divide by $p(P)$: $p(H, T | P) = p(H | P)p(T | P)$.

Step 11. Find $p(P = 1 | T = 1, H = 1)$.

Use the product from Step 10 and the values from Step 7:

$$\begin{aligned} p(P = 1, H = 1, T = 1) &= 0.2 \cdot 0.9 \cdot 0.92282 = 0.166108, \\ p(P = 0, H = 1, T = 1) &= 0.8 \cdot 0.2 \cdot 0.6647 = 0.106352. \end{aligned}$$

Sum them over P : $p(H = 1, T = 1) = 0.166108 + 0.106352 = 0.27246$. Divide:

$$p(P = 1 | T = 1, H = 1) = \frac{p(P = 1, H = 1, T = 1)}{p(H = 1, T = 1)} = \frac{0.166108}{0.27246} \approx 0.610.$$

Conclusion. The prior probability of the party is 0.2. A headache increases it to 0.529. An unhappy teacher increases it only to 0.258, because a bad answer has other causes, and the teacher is often unhappy without a bad answer. The two observations together give the highest probability, 0.610.

3. Let $x_1 \sim \text{Poiss}(\lambda_1)$ and $x_2 \sim \text{Poiss}(\lambda_2)$ be two independent Poisson random variables, that is, $p(x_i = k) = \exp(-\lambda_i) \frac{\lambda_i^k}{k!}$, $k = 0, 1, 2, \dots$. Prove that $x_1 + x_2 \sim \text{Poiss}(\lambda_1 + \lambda_2)$.

Idea. To find the distribution of a function $y = f(x_1, \dots, x_n)$ of discrete random variables, find $p(y = k)$ for each value k :

- (a) Write the event $\{f(x_1, \dots, x_n) = k\}$ as a union of disjoint events $\{x_1 = j_1, \dots, x_n = j_n\}$, one for each tuple with $f(j_1, \dots, j_n) = k$.
- (b) The probability of a union of disjoint events is the sum of their probabilities:

$$p(y = k) = \sum_{f(j_1, \dots, j_n) = k} p(x_1 = j_1, \dots, x_n = j_n).$$

- (c) If the variables are independent, write each joint probability as a product.
- (d) Simplify the sum and compare it with known distributions.

Solution.

Step 1. Write the event as a union of disjoint events.

The variables take only the values $0, 1, 2, \dots$. Thus $x_1 + x_2 = k$ if and only if $x_1 = j$ and $x_2 = k - j$ for one $j \in \{0, \dots, k\}$:

$$\{x_1 + x_2 = k\} = \bigcup_{j=0}^k \{x_1 = j, x_2 = k - j\}.$$

The events are disjoint, because each event has a different value of x_1 .

Step 2. Sum the probabilities of the disjoint events.

$$p(x_1 + x_2 = k) = \sum_{j=0}^k p(x_1 = j, x_2 = k - j).$$

Step 3. Use the independence of x_1 and x_2 .

$$p(x_1 + x_2 = k) = \sum_{j=0}^k p(x_1 = j) p(x_2 = k - j).$$

Step 4. Substitute the Poisson probabilities.

$$p(x_1 + x_2 = k) = \sum_{j=0}^k e^{-\lambda_1} \frac{\lambda_1^j}{j!} \cdot e^{-\lambda_2} \frac{\lambda_2^{k-j}}{(k-j)!}.$$

Step 5. Move the factors that do not depend on j out of the sum.

$$p(x_1 + x_2 = k) = e^{-\lambda_1} e^{-\lambda_2} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!} = e^{-(\lambda_1 + \lambda_2)} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!}.$$

Step 6. Multiply and divide by $k!$.

$$p(x_1 + x_2 = k) = \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \frac{k!}{j! (k-j)!} \lambda_1^j \lambda_2^{k-j} = \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \binom{k}{j} \lambda_1^j \lambda_2^{k-j}.$$

Step 7. Use the binomial theorem.

The binomial theorem gives $\sum_{j=0}^k \binom{k}{j} \lambda_1^j \lambda_2^{k-j} = (\lambda_1 + \lambda_2)^k$. Thus

$$p(x_1 + x_2 = k) = e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!}.$$

This is the probability mass function of $\text{Pois}(\lambda_1 + \lambda_2)$.

Conclusion. The sum of two independent Poisson random variables is a Poisson random variable, and the parameters add. By induction, the same is true for any finite number of independent Poisson random variables.

4. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Pois}(\lambda)$. Find the maximum likelihood estimate of λ .

Idea. Use the standard method for a maximum likelihood estimate:

- (a) Write the likelihood: the probability of the sample as a function of the parameter. For an independent sample, it is a product: $L(\lambda) = \prod_i p(x_i | \lambda)$.
- (b) Take the logarithm: $\ell(\lambda) = \log L(\lambda) = \sum_i \log p(x_i | \lambda)$. The logarithm is an increasing function, so the maximum point does not change.
- (c) Set the derivative to zero, $\ell'(\lambda) = 0$, and solve for the parameter.
- (d) Make sure that the solution is a maximum, for example with the sign of the second derivative $\ell''(\lambda)$.

Solution. Let $S = \sum_{i=1}^N x_i$. Then the sample mean is $\bar{x} = S/N$.

Step 1. Write the likelihood.

$$L(\lambda) = \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!}.$$

Step 2. Take the logarithm.

The logarithm of a product is the sum of the logarithms:

$$\ell(\lambda) = \sum_{i=1}^N \log \left(e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \right) = \sum_{i=1}^N (-\lambda + x_i \log \lambda - \log x_i!).$$

Step 3. Calculate each sum.

$$\ell(\lambda) = -N\lambda + S \log \lambda - \sum_{i=1}^N \log x_i!.$$

Step 4. Calculate the derivative.

The last term does not depend on λ :

$$\ell'(\lambda) = -N + \frac{S}{\lambda}.$$

Step 5. Set the derivative to zero and solve.

$$-N + \frac{S}{\lambda} = 0 \implies \frac{S}{\lambda} = N \implies \lambda = \frac{S}{N} = \bar{x}.$$

Step 6. Make sure that the solution is a maximum.

Calculate the second derivative:

$$\ell''(\lambda) = -\frac{S}{\lambda^2}.$$

If $S > 0$, then $\ell''(\lambda) < 0$ for all $\lambda > 0$. Thus ℓ is strictly concave, and the solution of Step 5 is the global maximum.

If $S = 0$, then all $x_i = 0$, and $\ell(\lambda) = -N\lambda$. This function decreases, so the maximum is at $\lambda = 0$. This value is also equal to $\bar{x}$.

The maximum likelihood estimate is the sample mean: $\hat{\lambda}_{\text{ML}} = \bar{x}$.

5. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. The prior distribution is the gamma distribution

$$p(\lambda) = \mathcal{G}(\lambda \mid a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda), \quad a, b > 0.$$

Find the Bayesian estimate of λ as the mean of the posterior distribution $p(\lambda \mid x_1, \dots, x_N)$.

Idea. Bayes' theorem gives the posterior distribution:

$$p(\lambda \mid X) = \frac{p(X \mid \lambda) p(\lambda)}{\int p(X \mid \lambda) p(\lambda) d\lambda}, \quad X = (x_1, \dots, x_N).$$

The numerator and the denominator contain the same function. The denominator only integrates it over λ . Thus you do not have to calculate the integral directly:

- (a) Write the numerator as $C \cdot g(\lambda)$. The constant C does not depend on λ .
- (b) Find a known distribution q with a density proportional to g : $g(\lambda) = Z q(\lambda)$, where Z is a known constant.
- (c) A density integrates to 1. Thus the denominator is

$$\int C Z q(\lambda) d\lambda = C Z \int q(\lambda) d\lambda = C Z.$$

- (d) Divide the numerator by the denominator:

$$p(\lambda \mid X) = \frac{C Z q(\lambda)}{C Z} = q(\lambda).$$

The same method gives the posterior mean without a direct integral.

Solution. Let $S = \sum_{i=1}^N x_i$.

Step 1. Write the likelihood as one product.

Collect the factors with $e^{-\lambda}$ and the powers of λ :

$$p(X \mid \lambda) = \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} = \frac{e^{-N\lambda} \lambda^S}{\prod_i x_i!}.$$

Step 2. Multiply the likelihood by the prior.

$$p(X \mid \lambda) p(\lambda) = \frac{e^{-N\lambda} \lambda^S}{\prod_i x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}.$$

Step 3. Separate the constant and the function of λ .

$$p(X \mid \lambda) p(\lambda) = \underbrace{\frac{b^a}{\Gamma(a) \prod_i x_i!}}_C \cdot \underbrace{\lambda^{a+S-1} e^{-(b+N)\lambda}}_{g(\lambda)}.$$

Step 4. Find the distribution with a density proportional to g .

Let $a' = a + S$ and $b' = b + N$. Then $g(\lambda) = \lambda^{a'-1} e^{-b'\lambda}$. The gamma density with these parameters is

$$\mathcal{G}(\lambda \mid a', b') = \frac{b'^{a'}}{\Gamma(a')} \lambda^{a'-1} e^{-b'\lambda} = \frac{b'^{a'}}{\Gamma(a')} g(\lambda).$$

Thus

$$g(\lambda) = \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda \mid a', b').$$

Step 5. Calculate the denominator.

Use Step 4. Then use the fact that the gamma density integrates to 1:

$$\int C g(\lambda) d\lambda = C \frac{\Gamma(a')}{b'^{a'}} \int \mathcal{G}(\lambda \mid a', b') d\lambda = C \frac{\Gamma(a')}{b'^{a'}}.$$

Step 6. Divide the numerator by the denominator.

$$p(\lambda \mid X) = \frac{C \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda \mid a', b')}{C \frac{\Gamma(a')}{b'^{a'}}} = \mathcal{G}(\lambda \mid a', b') = \mathcal{G}(\lambda \mid a + S, b + N).$$

Step 7. Write the posterior mean as an integral.

Substitute the gamma density:

$$\mathbb{E}[\lambda \mid X] = \int \lambda \mathcal{G}(\lambda \mid a', b') d\lambda = \frac{b'^{a'}}{\Gamma(a')} \int \lambda^{a'} e^{-b'\lambda} d\lambda.$$

Step 8. Calculate the integral with the same method.

The function under the integral is proportional to a gamma density with the parameters $a' + 1$ and b' :

$$\lambda^{a'} e^{-b'\lambda} = \lambda^{(a'+1)-1} e^{-b'\lambda} = \frac{\Gamma(a' + 1)}{b'^{a'+1}} \mathcal{G}(\lambda \mid a' + 1, b').$$

The gamma density integrates to 1. Thus

$$\int \lambda^{a'} e^{-b'\lambda} d\lambda = \frac{\Gamma(a' + 1)}{b'^{a'+1}}.$$

Step 9. Simplify the posterior mean.

Substitute Step 8 into Step 7. Then use $\Gamma(a' + 1) = a' \Gamma(a')$:

$$\mathbb{E}[\lambda \mid X] = \frac{b'^{a'}}{\Gamma(a')} \cdot \frac{\Gamma(a' + 1)}{b'^{a'+1}} = \frac{\Gamma(a' + 1)}{\Gamma(a') b'} = \frac{a' \Gamma(a')}{\Gamma(a') b'} = \frac{a'}{b'}.$$

The Bayesian estimate is

$$\hat{\lambda}_B = \frac{a'}{b'} = \frac{a + S}{b + N}.$$

Conclusion.

- The gamma prior and the Poisson likelihood give a gamma posterior. A prior with this property is *conjugate* to the likelihood. With a conjugate prior, the update only changes the parameters: $a \rightarrow a + S$ and $b \rightarrow b + N$. Thus a acts as a prior number of events, and b acts as a prior number of observations.

- The estimate is a weighted mean of the prior mean a/b and the maximum likelihood estimate $\bar{x} = S/N$ from Problem 4:

$$\frac{a + S}{b + N} = \frac{a}{b + N} + \frac{S}{b + N} = \frac{b}{b + N} \cdot \frac{a}{b} + \frac{N}{b + N} \cdot \frac{S}{N}.$$

When N increases, the weight $N/(b + N)$ of the data goes to 1, and the estimate goes to $\bar{x}$.

- When $a \rightarrow 0$ and $b \rightarrow 0$, the prior density is proportional to $\lambda^{a-1}e^{-b\lambda} \rightarrow 1/\lambda$. This is the non-informative prior from the notes below. With this prior, the Bayesian estimate is $(0 + S)/(0 + N) = \bar{x}$.

Notes on comparing the estimates from Problems 4 and 5.

1. For more about non-informative priors, see Section 5.4 “Priors” in Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*.
2. The prior $p(\lambda) \propto \frac{1}{\lambda}$ is a uniform prior on $\log \lambda$. To see this, use the change-of-variables rule for the density of a function of a random variable. See, for example, [this article](#).